

Vision Document

Smart HR Operations & Workforce Management System

Course Instructor

Ms. Laiba Imran

Submitted by

Group Number: 16

NAME	ROLL NUMBER
Sohaib Akhlaq	24i-3108
M. Hasaam	24i-3107
Shaiman Qasir	24i-3074

27-February-2026

Spring 2026



Department of Computer Science

FAST – National University of Computer & Emerging Sciences

Table of Contents

1. Document Version.....	3
2. Introduction.....	3
2.1 Purpose	3
2.2 Scope	3
2.3 Definitions, Acronyms, and Abbreviations	4
2.4 References	4
3. Business Opportunity.....	4
3.1 Business Case	5
3.2 Target Audience	5
4. Problem Statement (Structured).....	6
5. Summary of Features	7
5.1 Database	7
5.2 Accounts	7
5.3 Core Functional Modules	7
5.3.1 Employee Management.....	7
5.3.2 Leave and Attendance Management	8
5.3.3 Performance and Compliance Management.....	8
6. Product Overview.....	8
6.1 Product Perspective	8
6.2 Summary of Capabilities	9
6.3 Product Position Statement.....	9
7. Stakeholders.....	10
7.1 User Stakeholders.....	10
7.2 Non-User Stakeholders.....	10
8. Dependencies and Constraints.....	11
9. Cost and Pricing.....	11
10. Licensing and Installation	12

1. Document Version

Version	Date	Description	Authors
V1.0	27-02-2026	Initial Vision Document	Sohaib, Hasaam, Shaiman

2. Introduction

2.1 Purpose

This Vision Document aims at establishing and explaining the high-level objectives, scope, stakeholders, business opportunity, and key features of the Smart HR Operations and Workforce Management System. This document forms the basis of future architectural design, requirements and implementation planning. It determines the fundamental issues that the system is to solve, the value that it will provide to the stakeholders, and the general product direction according to the best practices of the software engineering.

2.2 Scope

The Smart HR Operations and Workforce Management System is a role-based and centralized enterprise solution that can be used to automate and control the human resource activities in organizations. It will be designed as a Java based application with the principles of the structured software architecture and the modeling techniques based on the UML.

The system will facilitate the major HR activities such as management of employee records, management of attendance, process of leaves, onboarding and offboarding processes, performance appraisal, and compliance reporting. It will give secure and role-based access to three main user groups, which include Admin, HR, and Employee. The platform will combine policy-based guidelines with workflow-based approval systems to create transparent operations which maintain consistent processes and achieve business efficiency.

2.3 Definitions, Acronyms, and Abbreviations

- a. HR – Human Resources
- b. RBAC – Role-Based Access Control
- c. Admin – System Administrator
- d. HRMS – Human Resource Management System

2.4 References

- a. SE2002 Course Guidelines

3. Business Opportunity

Managing HR operations are done in organizations especially in small and medium enterprises using manual processes, spreadsheets, or fragmented tools. Such strategies create inefficiencies, slowness in approvals, data discrepancies, absence of transparency and greater chances of policy breach. The need to have systematic and regulation-oriented HR management systems in which workflows are automated, organizational policies enforced and centralized views into workforce activities is on the increase.

The Smart HR Operations & Workforce Management System fits this gap by providing a policy-enforced and configurable and scaled HR solution designed to support the modern organization. The system offers organizations better operational control, less administration overhead and better compliance tracking as well as increased employee engagement via self-service.

3.1 Business Case

The Smart HR Operations & Workforce Management System is a system that offers quantifiable organizational value through the minimization of administrative overhead, policy violations, the enhancement of approval efficiency, and the increase in workforce transparency. The system will decrease the amount of manual effort and operational expenses by automating repetitive HR operations, enhance compliance and managerial decision-making abilities. The system can also be provided as an enterprise solution, in a subscription-based system, which is customized to the size of the organization and it ultimately creates recurrent revenue streams.

3.2 Target Audience

The major target audience would be:

- Small and Medium Enterprises (SMEs)
- Growing startups with structured HR needs
- Educational institutions managing staff records
- Organizations transitioning from manual HR processes to automated systems

The secondary audience will involve HR managers, administrative heads as well as operational directors in need of better governance and workforce analytics.

4. Problem Statement

Element	Description
The problem of	inefficient, manual and fragmented HR processes due to the disconnected nature of attendance systems, spreadsheet-based employee records, and informal approval processes.
Affects	Small and medium organizations, HR departments, managers and employees who manage workforce through non centralized systems.
The impact of which is	Delayed approvals, inaccurate employee information, inconsistency of policies, lack of transparency, low operational productivity, risk of compliance and insufficient managerial understanding of workforce performance.
A successful solution would be	A central, safe, and rule-based HR management that automates working processes, implements flexible HR policies, offers chain of approval, enables role-based access control and creates real-time compliance and analytics reports.

5. Summary of Features

5.1 Database

The system shall support a central relational database which will store all employee records and attendance records and leave balance information and performance assessments and department organizational data and system setup details. The database needs to maintain data integrity while providing consistent data access and enabling users to perform auditing procedures. The system will maintain historical records for both reporting needs and compliance requirements.

5.2 Accounts

The system shall support secure account management using role-based access control. Users shall be assigned predefined roles including Admin, HR, and Employee. Each role shall have restricted and well-defined permissions. Authentication shall be mandatory for system access, and sensitive operations shall be logged for accountability.

5.3 Core Functional Modules

5.3.1 Employee Management

The system shall allow HR to manage employee onboarding and offboarding workflows. The system will store employment status information together with departmental assignments, designation records, probation tracking data, and role allocation details. Admin shall define departments and organizational structure.

5.3.2 Leave and Attendance Management

The system shall allow employees to submit leave requests and attendance correction requests. It shall automatically validate requests based on configured HR policies. The system shall enforce rule-based approvals through HR and Admin workflows. Attendance summaries and leave balances shall be generated dynamically.

5.3.3 Performance and Compliance Management

The system needs to provide support for three functions which include performance tracking and evaluation cycles and probation monitoring. HR shall evaluate employees using structured criteria. The system will create three types of reports which include compliance reports and attendance summaries and organization-wide HR analytics reports for administrative decision-making.

6. Product Overview

6.1 Product Perspective

The Smart HR Operations & Workforce Management System functions as an enterprise application that uses multiple software architecture layers to create its centralized system. The system has three operational layers which include presentation components and business logic functions and data management systems. The system enforces operational processes through workflow execution together with its rule-based validation system. The system architecture provides three main benefits which include system expansion and ongoing software maintenance and protection against security threats while it supports upcoming payroll system links and external business system connections.

6.2 Summary of Capabilities

- i. The system provides centralized employee record management.
- ii. The system supports automated leave and attendance workflows.
- iii. The system enforces HR policies through configurable rule engines.
- iv. The system offers formal onboarding and offboarding processes.
- v. The system enables performance tracking and probation evaluation.
- vi. The system produces reports and analytics that are compliance-ready.
- vii. The system has secured access by role-based permissions.

6.3 Product Position Statement

Element	Description
For	Small and medium-sized organizations that need organized and transparent HR operations.
Who	Requires automating the performance of employees, implementing HR policies, simplifying processes of approvals, and enhancing visibility of workforce.
The Smart HR Operations & Workforce Management System	Is an automated HR platform that is a role-based and centralized system.
That	Provides policy-based validation, workflow-based approvals, centralized employee records, real-time attendance tracking, performance tracking and compliance reporting.
Unlike	Traditional spreadsheet-based systems or simple HR applications that merely archive the data of their employees without imposing any structured workflow or customizable policy regulations.
Our product	Imposes decision logic and approval hierarchy governed by rules, audit tracking and customizable HR policies that are designed to assure operational efficiency, visibility, compliance and accountability.

7. Stakeholders

7.1 User Stakeholders

- i. Admin will be assigned with the duties of setting up system policies, defining the departments, controlling roles and producing compliance reports. Admin makes sure that the system parameters are in line with organizational rules.
- ii. The HR handles various lifecycle operations of employees such as onboarding, leave approvals, monitoring employee attendance and also carry out performance appraisals. HR provides stability of operations and employee control.
- iii. Employee is responsible for managing personal HR interactions including leave applications, attendance corrections, and viewing personal records and performance history.

7.2 Non-User Stakeholders

- i. System Analyst collects the needs of the stakeholders and outlines system scope.
- ii. Software Architect determines the system architecture, scalability and architectural integrity.
- iii. Project Manager organizes and monitors the development milestones and resources.
- iv. Organization management uses review reports and analytics in strategic workforce choices.
- v. Regulatory compliance Generated reports may be reviewed by Compliance Authorities.

8. Dependencies and Constraints

- i. The system relies on the effective database management system and security authentication.
- ii. The system will be created based on Java technologies and comply with layered architectural patterns.
- iii. The security restrictions comprise of mandatory authentication, encrypted password storage, role-based authorization, and audit logging.
- iv. Performance constraints require that routine operations such as leave approval and attendance viewing respond within acceptable system thresholds under normal load conditions.
- v. The system shall operate within the scope defined for academic implementation and shall not include payroll integration in the initial release.

9. Cost and Pricing

Item	No. of Items	Cost	Total
Developers	3	40,000	120,000
Development Tools	1	15,000	15,000
Database setup	1	20,000	20,000
Others (Testing and Miscellaneous)	-	10,000	10,000
Total	-	-	165,000 PKR

The academic prototype version is developed for educational purposes. In a commercial setting, pricing would follow a subscription-based licensing model depending on organization size and number of active users.

10. Licensing and Installation

The Smart HR Operations & Workforce Management System will be deployed as a centralized enterprise application. Installation shall require configuration of database services and server runtime environment. The academic version is intended for educational and demonstration purposes only. A commercial version would require organizational licensing agreements and structured deployment procedure.